Database System

<u>Assignment# 03</u>

University of Rasul, Mandi Bahaud Din

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Database Normalization

Question:

A university stores student enrollment data in a single table:

- Student_ID
- Student_Name
- Course_ID
- Course_Name
- Instructor_Name
- Instructor_Office
- Department
- Department_HOD

A student can enroll in multiple courses, and one instructor can teach multiple courses. Each department has only one HOD..

Problems in the Unnormalized Table

The given table contains several problems:

a) Data Redundancy

The same student, course, instructor, and department data is repeated many times.

Student_ID	Student Name	Course_ID	Course_Name
101	Ali	C01	Database
101	Ali	C02	OOP
103	Usman	C03	DLD

b) Update Anomaly

If the HOD changes, we must update it in many rows.

Example: If Computer Science HOD changes from "Dr. Ahmad" to "Dr. Khan", every row must be updated.

c) Insertion Anomaly

We cannot insert a department unless a student is enrolled.

Example: If a new department starts but no student is registered yet, its data cannot be stored properly.

d) Deletion Anomaly

If the last student of a course is deleted, course information is also lost.

Example: Deleting the last student enrolled in "Database" may remove the course details too.

Conversion to First Normal Form (1NF)

Rule of 1NF

- No repeating groups
- Each column contains atomic (single) values

1NF Table:

Student_ID	Student_Name	Course_ID	Course_Name	Instructor_Name	Instructor_Office	Department	Department_HOD
S101	Ali	C01	Database	Sir Irfan	Room 12	CS	Dr. Bilal
S101	Ali	C02	OOP	Dr. Sana	Room 15	CS	Dr. Bilal
S102	Ayesha	C01	Database	Sir Irfan	Room 12	CS	Dr. Bilal

Conversion to Second Normal Form (2NF)

Rule of 2NF:

- Table must already be in 1NF
- No partial dependency

Problem

The primary key is:

(Student_ID, Course_ID)

But:

- Student_Name depends only on Student_ID
- Course_Name depends only on Course_ID
- Instructor details depend on Course_ID

So we divide the table**.**

Student Table:

Student_id	**Student_Name**
101	Ali
102	Ayesha

Course Table:

Course_id	**Course_Name**	**Ins_name**	**Ins_office**	**Dept**	**Dept_hod**
101	Database	Sir. irfan	Room 12	IT	Dr. bilal
102	OOP	Mam sana	Room 15	IT	Dr. bilal

ENROLLMENT Table:

Student_id (PK)	**Course_id (PK)**
101	C01
101	C02
102	C01

Now partial dependencies are removed.

Conversion to Third Normal Form (3NF)

Rule of 3NF:

- Table must already be in 2NF
- No transitive dependency

Problem:

In COURSE table:

- Department → Department_HOD
- Department_HOD does not depend directly on Course_ID.

So we create a separate DEPARTMENT table.

Final Tables in 3NF

STUDENT Table:

Student_id(PK)	**Studen tname**
101	**Ali**
101	**Ayesha**

Department table:

Dept_id	Dept_Name	Dept_HOD
D01	IT	Dr. bilal

INSTRUCTOR Table:

INS_ID	INS_NAME	INS_OFFICE	DEPT
101	Sir.irfan	Room 12	D01
101	Mam sana	Room 15	D01

Course Table:

Course_id	Course_name	Ins_id
C01	Database	101
C02	OOP	101

ENROLLMENT Table:

Student_id	Course_id
101	C01
101	C02
102	C01

Primary Keys and Foreign Keys

Primary Keys (PK)

- STUDENT → Student_ID
- DEPARTMENT → Department_ID
- INSTRUCTOR → Instructor_ID
- COURSE → Course_ID
- ENROLLMENT → (Student_ID, Course_ID)

Foreign Keys (FK)

- INSTRUCTOR.Department_ID → DEPARTMENT.Department_ID
- COURSE.Instructor_ID → INSTRUCTOR.Instructor_ID
- ENROLLMENT.Student_ID → STUDENT.Student_ID
- ENROLLMENT.Course_ID → COURSE.Course_ID

Benefits of Normalization

a) Reduces Data Redundancy

Duplicate data is minimized.

Example: Student information is stored only once.

b) Reduces Update Anomalies

Changes are made in one place only.

Example: Changing HOD name requires updating only the DEPARTMENT table.

c) Reduces Insertion Anomalies

New departments, instructors, or courses can be added independently.

d) Reduces Deletion Anomalies

Deleting one record does not remove important unrelated data.

Example: Deleting a student enrollment will not delete course details.

Conclusion:

Normalization organizes the database into smaller related tables.

It improves:

- Data consistency
- Data integrity
- Storage efficiency
- Database management

The final database is successfully converted into Third Normal Form (3NF).